

CITY MULTI Compressor Replacement Instructions

Purpose

These instructions are intended only for and be utilized by a licensed HVAC professional skilled and experienced with compressor replacement and refrigerant related repairs using industry standard tools and equipment. They are not intended for a do it yourself. Attempted use of these instructions by anyone but an experience, licensed HVAC professional may result in property damage, fire, accident and injury can occur such as burns, asphyxiation, cuts, and other bodily injury including death. Before proceeding' be fully confident in your ability to perform such procedures. If you feel that these procedures are beyond your capabilities please consult another HVAC professional or your nearest Mitsubishi dealer or representative. Mitsubishi Electric Trane HVAC US LLC Inc. makes no representations or warranties regarding these instructions and the installer assumes all risks in their utilization.

Procedure

- 1) Check and confirm power to the unit is off and tagged out.
- 2) Remove exterior panels necessary to access the refrigerant access ports, compressor, and accumulator.
- 3) Recover the refrigerant charge from both service ports on the high and low side: Figure 1 - Note current models do not have the L shaped adapters and you will have to use the fittings on the valves. It may also be necessary to apply heat to the accumulators and compressors to insure that all the refrigerant is recovered. Caution open flames should not be used as a heat source to heat the accumulator or compressor.

Figure 1



- 4) Remove the access cover from the compressor compartment: Figure 2 and 3

Figure 2



Figure 3



- 5) Remove the top cover to the compressor box. Figure 4 - Depending on model you may have to remove the blanking cover above the compressor compartment and beside the electrical panel. Figure 5 - Some models may have a transformer box on top of the compressor box that will need to be removed in order to remove the top cover. Figure 6 - If you are replacing the accumulator and the model you have has the blanking plate it will also be necessary that the electrical panel be lowered to make removing the accumulator possible. Figure 7

Figure 4

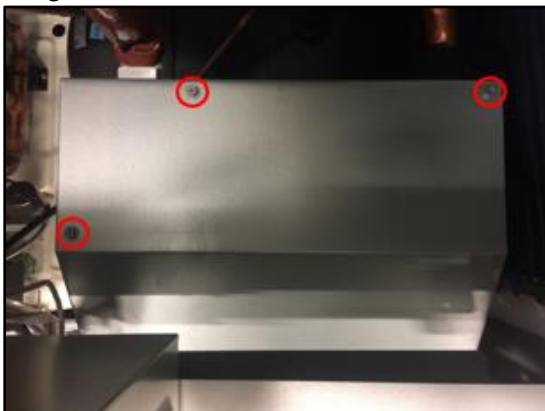


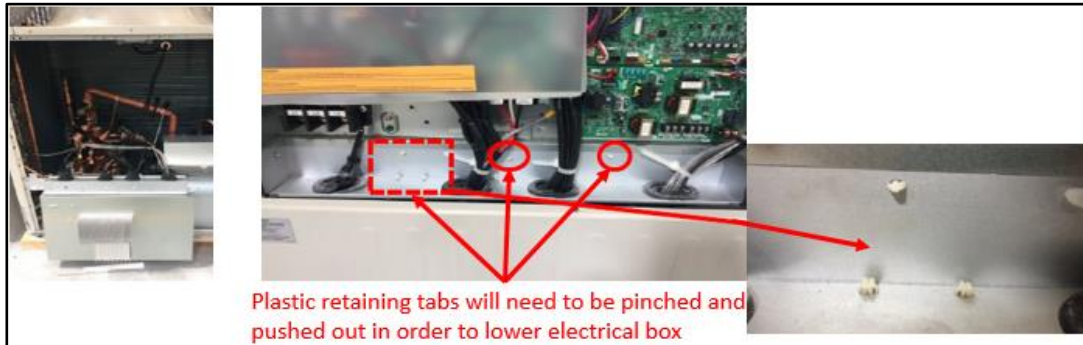
Figure 5



Figure 6



Figure 7



- 6) De-braze the suction line joint and the discharge line joint closest to the compressor. Figure 8 – If you are replacing the accumulator de-braze the two joints closest to the accumulator.

Figure 8



- 7) Remove the compressor mounting bolt nuts and the anti-vibration bracket. Figure 9 - Note that there will be one nut missing. This is normal. If there is a crankcase heater and or a thermistor strapped on the bottom of the compressor remove them. The thermistor will be secured with a stainless steel strap and taped and glued to the compressor. To break the glue bond use a chisel or screw driver and tap at the point at which the thermistor is glued to the compressor. Figure 10 - If the accumulator is being replaced remove the four bolts holding it. Figures 11 - If there is a liquid level detection belt on the accumulator (Figure 12) please consult a Mitsubishi representative about replacing this component.

Figure 9



Figure 10

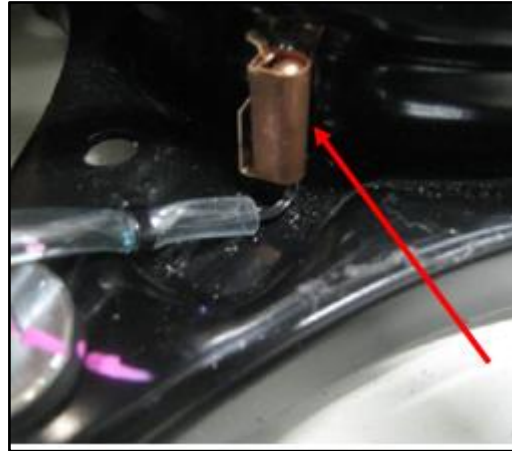


Figure 11

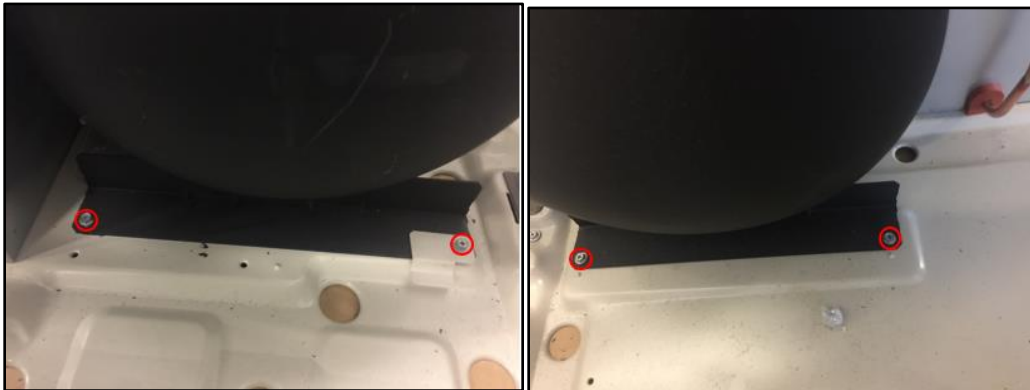
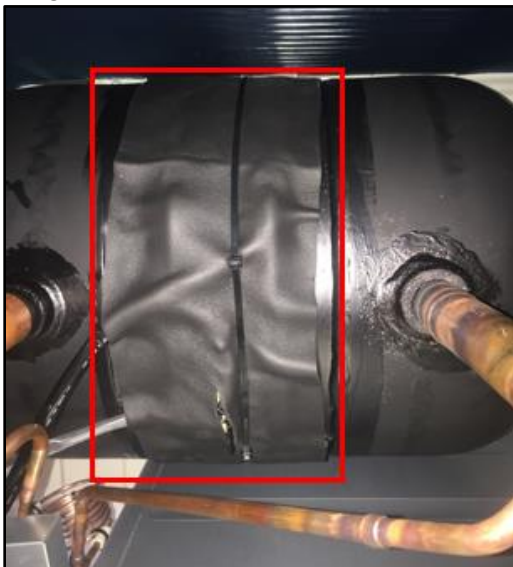
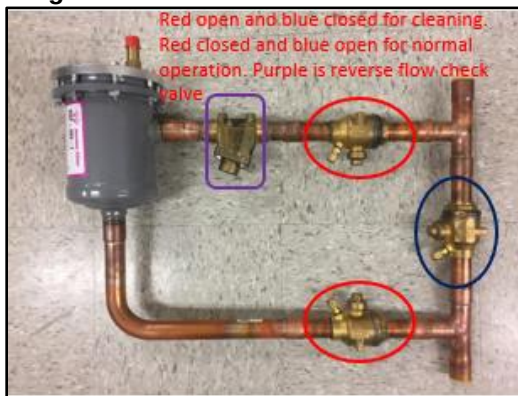


Figure 12



- 8) Remove the compressor. If you are replacing the accumulator remove it also at this point.
- 9) Tilt and pour the oil from the compressor suction pipe into a 3 liter container and observe the quantity of oil. Also see if there is any foreign matter in the oil. If you have less than one liter or 32 ounces of oil please consult with a Mitsubishi representative. If there is foreign matter in the oil consult with a Mitsubishi representative.
- 10) Perform an acid test on the oil to see if and how much acid is present. If you are installing the PAC-SPRFCS-*** filter kit (Figure 13) this will help to determine the type of filter needed in the filter kit. If there is acid use the solid core filter part RC-4864-HH (404380)

Figure 13



- 11) Install the new accumulator and compressor.
- 12) If you are installing the PAC-SPRFCS-*** filter kit install it in the common suction line per instructions.
- 13) Braze all braze joints while purging the system with nitrogen at a pressure of .5 to 2 PSI
- 14) Install the compressor mounting bolt nuts and anti-vibration bracket. Install the thermistor on the bottom of the compressor if so equipped using a strip of silver tape to hold in place and then placing the stainless strap on top Figure 14. - Install the crankcase heater if equipped. If replacing the accumulator install the mounting bolts.

Figure 14



- 15) Pressurize the system with nitrogen at the ball valves and verify that the braze joints do not leak. Recommended pressure of 600PSI.
- 16) Release the nitrogen pressure and evacuate the system to 4000 microns
- 17) Break the vacuum to 0 PSI using nitrogen and re evacuate the system to 1500 microns
- 18) Break the vacuum to 0 PSI using nitrogen and re evacuate the system to 500 microns
- 19) Once the vacuum reaches 500 microns let the pump run for 30 minutes and then close the gauge valves.
- 20) Shut the pump off and allow the system to stand for 1 hour and make sure the vacuum remains at 500 microns.
- 21) Weigh the appropriate amount of charge back into the system. Only charge liquid into the high side of the system and not into the low side. If you installed a new accumulator it requires that the oil charge for the accumulator be added to the system. Prior to charging add the oil into the suction line before the accumulator. The oil container has two service valves with gauge ports at each. Figure 15 (**PN - R63 Y01 621**) - Stand the oil tank on its stand in a level position. Attach the bottom port of the tank to the low pressure side of the system. Service port just below the electrical panel. Figure 16 - Attach the top of the tank to the refrigerant cylinder or the high side of the system if charge has already been added. Purge both refrigerant lines and open the service valves with higher pressure at the top. This will force the oil out of the bottom of the tank and into the system. When the tank is empty close the valves.

Figure 15

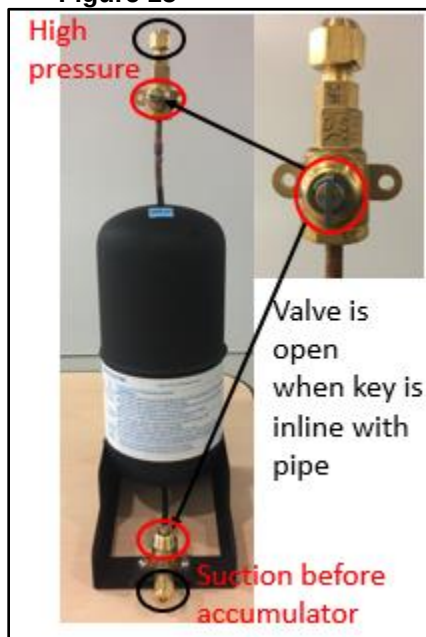
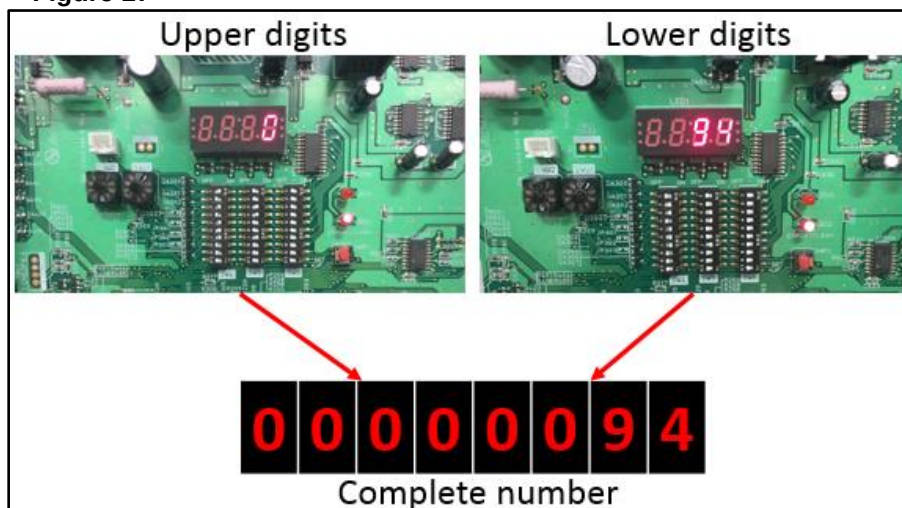


Figure 16



- 22) Check the compressor operation hours and record on the electrical panel the number of hours and the date that the compressor is replaced. Operation hours can be obtained by dip switch settings. For J generation units turn on SW1-2, 3, 5, 6, and 7 for lower digits and SW1-1, 3, 5, 6, and 7 for upper digits. For K and L generation units turn on SW4-2, 3, 5, 6, and 7 for the lower 4 digits and SW4-1, 3, 5, 6, and 7 for the upper 4 digits. Hours are recorded as upper digits higher and lower digits lower. Figure 17

Figure 17



- 23) Reset the compressor operation hours to zero. On J gen models turn on SW4-6 while the unit is powered and wait one minute and then turn back off. On K and L gen models turn on dip switches SW4-7, 9, and 10 and SW6-10. 832 should display on the screen. Figure 18 - Now push the SWP1 button to the right of the dip switches and hold it until LED3 flashes on. Now set the switches back and confirm that the hours are back to zero using the SW4 settings from step 22.

Figure 18



- 24) Replace all panels.
- 25) If you installed the PAC-SPRFCS-*** filter kit and had acid in the system perform a system cleaning per instructions that came with the filter kit.